

USB/OTG Review (TC53e + RFD40 - Kotlin Edition)

Scope

This review covers USB-related RFID connection and disconnection behavior under various device modes (charging-only, file transfer, debug mode, and general RFID operations) in: - app/src/main/java/com/zebra/rfid/demo/sdksample/MainActivity.kt - app/src/main/java/com/zebra/rfid/demo/sdksample/RFIDHandler.kt

1. Gating & USB Mode Detection

The application implements active USB broadcast and state monitoring. It listens to three key intents registered in MainActivity.kt: - android.hardware.usb.action.USB_STATE (ACTION_USB_STATE) - android.intent.action.ACTION_POWER_CONNECTED (ACTION_POWER_CONNECTED) - android.intent.action.ACTION_POWER_DISCONNECTED (ACTION_POWER_DISCONNECTED)

Mode Determination logic

USB features are decoded in updateUsbModeFromIntent(Intent, String) where the system parses the following boolean intent extras: - connected / configured - Explicit transport functions: mtp, ptp, mass_storage, adb, rndis

The state flag usbFileTransferModeActive is used to decide whether to trigger immediate disconnection: - **Interactive / Data Link Mode (File Transfer / Debug)**: Activated if any transport function (mtp, ptp, mass_storage, adb, rndis) is true. - **Fallback Rule**: In custom operating system builds where individual function extras are not broadcasted, the fall-back checks if the USB configuration is both **connected** and **configured** to infer an active data connection. - **Power-Only (Charging-Only)**: Inferred when ACTION_POWER_CONNECTED is fired but usbFileTransferModeActive is false.

2. Mode-Specific Connection & Disconnection Behavior

2.1 Power-Only Gating (Pass-through Charger)

- **Design Goal**: Disconnect from the active session on a shared-USB reader during charging and allow pass-through charging. Skip selecting the host reader device while powered.
- **On Intent (ACTION_POWER_CONNECTED)**:
 1. Activates host suppression: rfidHandler?.setSkipTc53eReaderSelection(true).
 2. Stops the reader session cleanly: rfidHandler?.onPause().
 3. Displays a warning toast: "RFID disconnected while USB power cable connected".
- **On Intent (ACTION_POWER_DISCONNECTED)**:
 1. Restores default search modes: rfidHandler?.setSkipTc53eReaderSelection(false).
 2. Orchestrates debounced reconnection attempts using a timed retry window (startPowerReconnectWindow()).

2.2 USB File Transfer / Debug Mode

- **Design Goal**: Retain the RFID session until the reader hardware physically disappears (authoritative detachment) during data transfers and software debugging.
- **On Intent (ACTION_POWER_CONNECTED)**:
 1. Preserves host-reader fallback flags: rfidHandler?.setSkipTc53eReaderSelection(false).
 2. Forgets previous power latches and warns the user that files are being shared/debugged, but does **not** command an automatic force-disconnect.
 3. Displays a warning status popup: "USB file transfer active or debug mode\r\nRFID Disconnected\r\nWait for USB cable unplug for reconnect".

2.3 Reader-Detached & Disappear Events

- **Design Goal:** Hard detachment and state cleanup are prioritized over power state checks when the hardware is decoupled.
- **On Event (RFIDReaderDisappeared):**
 1. Authoritatively registers state: `detachedEventInProgress = true`.
 2. Immediately proceeds with state cleanup in `RFIDHandler.kt`:
 - Unregisters the event listener from `Reader.Events`.
 - Safely closes connection transport handled by `reader.disconnect()`.
 - Clears the instance: `reader = null` and resets `readersAttached = false` inside `finally` block to allow fresh bindings in the next lifecycle.
 3. UI warning text: "Reader detached: <name>". The generic disconnect toast is suppressed to avoid redundant status noise.

2.4 TC22R Bypass

- **Design Goal:** Guard standard TC22R operations against accidental USB-state interruptions.
 - **Logic Gating:** Evaluated via `rfidHandler?.isTC22R()`. If `true`, the broadcast receiver bypasses the entire USB disconnect policy block, outputs `STEP: TC22R`, ignore `ACTION_POWER_CONNECTED`, and confirms active operational status directly on the interface.
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3. Reconnect Retries and Suppression Window

To handle immediate transport races that can occur right after the cable is unplugged, the app implements a bounded retry schedule in `MainActivity.kt`:

1. **Backoff Delays:** Attempts are staggered at 500 ms → 1200 ms → 2500 ms to allow port states and physical interfaces to settle down.
 2. **Busy Checks:** Before executing `requestReaderResumeDebounce()`, the handler evaluates `rfidHandler?.isConnectionBusy()`. If initialization or another connection attempt is active, it reschedules itself after a short 400 ms delay instead of causing concurrent thread calls.
 3. **UX Suppression:** During this 11,000 ms window, intermediate transport failures (e.g., `RFID_COMM_OPEN_ERROR`) are hidden from the user interface.
 4. **Outcome Hook:**
 - **Success:** If a loop succeeds and reports a status containing "connected", the reconnect window is immediately closed, and suppression state is cleared.
 - **Timeout:** If the window times out, it displays: "USB in used!!!\r\nUnplug the USB cable for reconnect".
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4. Current State Matrix

Feature / Scenario	Mode Decided	Skip Selection Flag	RFID Connect Policy	Expected UI Status / Toast
AC Charger Connected	Charging Only	true	Call <code>onPause()</code> , Disconnect	RFID disconnected while USB power cable connected
USB Data Connected	File Transfer / Debug	false	Keep active session	USB file transfer active or debug mode...
TC22R Connected	Any USB	false	Bypass disconnect	Connected to TC22R

Feature / Scenario	Mode Decided	Skip Selection Flag	RFID Connect Policy	Expected UI Status / Toast
Charger Unplugged	Power Disconnected	false	Debounced Retry Window	USB unplugged. Reader Connected (<i>on success</i>)
Retry Window Timeout	Persistent Timeout	false	End Retries	USB in used!!!\r\nUnplug the USB cable for reconnect
Physical Sled Unplug	Hardware Detach	false	Call disconnect(), Reset state	Reader detached: <name>

5. Suggestions for Code Validation

- ☐ **Pass-through Charging Verification:** Plug into a standard USB charging adapter. Ensure current reader session is dropped immediately and status reflects power disconnection guidelines.
- ☐ **Interactive Mode Gating:** Plug into a development machine and verify that debug session logs output details from `updateUsbModeFromIntent` indicating explicit data-mode transport functions. Ensure RFID operations can continue until the reader is physically detached.
- ☐ **Debounced Recovery:** Rapidly plug and unplug the cable to verify that the latching logic and debouncing timers correctly drop duplicate requests and execute sequentially.
- ☐ **TC22R Verification:** Verify that a TC22R handheld device does not undergo accidental power-connected drop events.